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Transfer Flow Statement:

III. Transfer Flow Statements

Purpose:

The purpose of Transfer Flow Statements is to change the control of the Python Virtual Machine (PVM) from one part of the program to another part of the program.

In Python programming, there are four types of Transfer Flow Statements:

1. `break` – Terminates the loop and transfers control outside the loop.
2. `continue` – Skips the current iteration and moves to the next iteration of the loop.
3. `pass` – A null statement; it does nothing but is used as a placeholder.
4. `return` – Exits from a function and optionally returns a value.

break :-

Break Statement

1. Break Statement

- `break` is a keyword in Python.
- The `break` keyword must be used inside loops (for or while), otherwise a Syntax Error occurs.
- The purpose of the `break` statement is to terminate the execution of a loop immediately when a certain condition is satisfied.
- When a `break` statement occurs inside a for loop or while loop, the Python Virtual Machine (PVM) stops the loop execution and exits from the

loop.

- If `break` executes inside a loop, the corresponding `else` block will not be executed.

```
In [27]: # Program for Demonstrating the Need of break Statement
s = "PYTHON"
print("By using for loop with break statement")
for ch in s:
    print(ch)
else:
    print("I am from out of for loop")
print("-----")
# To display the index without using indexing and slicing
for ch in s: # PYTHON
    if(ch=="O"):
        break
    print(ch, end=" ")
else:
```

```

    print("I am from out of for loop")
print()
print("-----")

```

By using for loop

P

Y

T

H

O

N

I am from out of for loop

PYTH

In [31]: *# Program for Demonstrating the Need of break Statement*

```

s = "PYTHON"

print("By using while loop")

i = 0
while (i < len(s)):
    print(s[i])
    i = i + 1
else:
    print("I am from out-off while loop")

print("-----")

# Today my requirement is to display "PYTH"
# without using indexing and slicing

print("By using while loop with break statement")

i = 0
while (i < len(s)):
    if (s[i] == "O"):
        break
    print(s[i], end="")
    i = i + 1
else:
    print("I am from out-off while loop")

print()
print("-----")

```

By using while loop

P

Y

T

H

O

N

I am from out-off while loop

By using while loop with break statement

PYTH

```
In [35]: #25:31
# Program for Demonstrating the Need of break Statement

s = "Mahaboob khan"

print("By using for loop with break statement")
print("-----")

for ch in s:
    if ch == "b":
        break
    print(ch, end="")

print()
print("-----")
```

By using for loop with break statement

Maha

```
In [37]: # Program for Demonstrating the Need of break Statement

s = "MRIIRS"

print("By using for loop with break statement")
print("-----")

for ch in s:
    if ch == "I":
        break
    print(ch, end="")

print()
print("-----")
```

By using for loop with break statement

MR

```
In [39]: #30:31
# Program for Demonstrating the Need of break Statement
```

```

s = "MISSISSIPPI"

# Display MISS
print("By using for loop with break statement")
print("-----")

cnt = 0

for ch in s:
    if ch == "I":
        cnt = cnt + 1

    if cnt == 2:
        break

    print(ch, end="")

print()
print("-----")

```

By using for loop with break statement

```

-----
MISS
-----

```

```

In [41]: # 30:22:
# Program for Demonstrating the Need of break Statement
# BreakStmtEx3.py

s = "MISSISSIPPI"

cnt = 0
for ch in s:
    if ch == "I":
        cnt = cnt + 1
    if cnt == 2:
        break
    print(ch, end="")

print()

print("----- OR -----")

print("By using for loop with break statement with enumerate()")
print("-----")

for index, ch in enumerate(s):
    print("\t{}\t\t{}".format(index, ch))

```

MISS

----- OR -----

By using for loop with break statement with enumerate()

0	M
1	I
2	S
3	S
4	I
5	S
6	S
7	I
8	P
9	P
10	I

```
In [43]: #45:00
# Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if (n <= 1):
    print("{} is Invalid Input".format(n))
else:
    for i in range(2, n):
        if (n % i == 0):
            print("NOT PRIME")
            break
    else:
        print("PRIME")
```

PRIME

```
In [45]: # Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if (n <= 1):
    print("{} is Invalid Input".format(n))
else:
    for i in range(2, n):
        if (n % i == 0):
            print("NOT PRIME")
            break
    else:
        print("PRIME")
```

1 is Invalid Input

```
In [47]: # Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if (n <= 1):
    print("{} is Invalid Input".format(n))
else:
    for i in range(2, n):
```

```

        if (n % i == 0):
            print("NOT PRIME")
            break
    else:
        print("PRIME")

```

0 is Invalid Input

```

In [49]: # Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if n <= 1:
    print("{} is Invalid Input".format(n))
else:
    res = "PRIME"
    for i in range(2, n):
        if n % i == 0:
            res = "NOT PRIME"
            break

    # out of for loop
    print("{} is {}".format(n, res))

```

5 is PRIME

```

In [51]: # Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if n <= 1:
    print("{} is Invalid Input".format(n))
else:
    res = "PRIME"
    for i in range(2, n):
        if n % i == 0:
            res = "NOT PRIME"
            break

    # out of for loop
    print("{} is {}".format(n, res))

```

8 is NOT PRIME

```

In [10]: # 54:04
# Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if n <= 1:
    print("{} is Invalid Input".format(n))
else:
    res = True
    for i in range(2, n):
        if n % i == 0:
            res = False
            break

```

```
# Out of for Loop
r = "Prime" if res else "NOT Prime"
print("{} is {}".format(n, r))
```

12 is NOT Prime

```
In [12]: # 54:04
# Program for deciding whether the given number is prime or not

n = int(input("Enter a Number: "))

if n <= 1:
    print("{} is Invalid Input".format(n))
else:
    res = True
    for i in range(2, n):
        if n % i == 0:
            res = False
            break

    # Out of for Loop
    r = "Prime" if res else "NOT Prime"
    print("{} is {}".format(n, r))
```

13 is Prime

```
In [16]: # 01:02:00 Find the uppercase Letters in given Line and Number of uppercase Letters
#Line of Text:
#PyTHon IS an OOp LaNg

#Uppercase Letters
#P T H I S O O L N

#Number of Uppercase Letters

# Program for accepting a line of text and counting the number of uppercase Letters
line = input("Enter Line of Text: ")

print("-----")
print("Given Line of Text: {}".format(line))
print("-----")

noup = 0

for value in line:
    if value.isupper():
        print(value, end=" ")
        noup = noup + 1

print()
print("Number of Upper Case Letters = {}".format(noup))
```



```
-----
Given Line of Text: PyTHon IS an 00p LaNg
-----
```

```
P T H I S O O L N
```

```
Number of Upper Case Letters = 9
```

```
In [18]: # Program for accepting a line of text and counting the number of uppercase Letters
line = input("Enter Line of Text: ")

print("-----")
print("Given Line of Text: {}".format(line))
print("-----")

noup = 0

for value in line:
    if value.isupper():
        print(value, end=" ")
        noup = noup + 1

print()
print("Number of Upper Case Letters = {}".format(noup))
```

```
-----
Given Line of Text: MaHaBooB KhAn
-----
```

```
M H B B K A
```

```
Number of Upper Case Letters = 6
```

```
In [20]: # Program for accepting a line of text and counting the number of uppercase Letters
line = input("Enter Line of Text: ")

print("-----")
print("Given Line of Text: {}".format(line))
print("-----")

noup = 0

for value in line:
    if value.isupper():
        print(value, end=" ")
        noup = noup + 1

print()
print("Number of Upper Case Letters = {}".format(noup))
```

```
-----
Given Line of Text: MrIiRs
-----
```

```
M I R
```

```
Number of Upper Case Letters = 3
```

```
In [22]: # 01:13:00FindSpecialSymbols.py

line = input("Enter Line of Text: ") # Example: PyTHon IS an 00p LaNg!

print("-----")
```

```

print("Given Line of Text: {}".format(line))
print("-----")

nosp = 0

for value in line:
    if (not value.isalnum()) and (not value.isspace()):
        print(value, end=" | ")
        nosp = nosp + 1

print()
print("Number of Special Symbols = {}".format(nosp))
print("-----")

```

```

-----
Given Line of Text: PyTHon IS an 00p LaNg!
-----
! |
Number of Special Symbols = 1
-----

```

In [24]: # 01:13:00FindSpecialSymbols.py

```

line = input("Enter Line of Text: ") # Example: MaHaBoOb KhAn@23

print("-----")
print("Given Line of Text: {}".format(line))
print("-----")

nosp = 0

for value in line:
    if (not value.isalnum()) and (not value.isspace()):
        print(value, end=" | ")
        nosp = nosp + 1

print()
print("Number of Special Symbols = {}".format(nosp))
print("-----")

```

```

-----
Given Line of Text: MaHaBoOb KhAn@23
-----
@ |
Number of Special Symbols = 1
-----

```

In [26]: # 01:13:00FindSpecialSymbols.py

```

line = input("Enter Line of Text: ") # Example: MrIirs@23

print("-----")
print("Given Line of Text: {}".format(line))
print("-----")

nosp = 0

```

```
for value in line:
    if (not value.isalnum()) and (not value.isspace()):
        print(value, end=" | ")
        nosp = nosp + 1

print()
print("Number of Special Symbols = {}".format(nosp))
print("-----")
```

Given Line of Text: MrIirs@23

@ |
Number of Special Symbols = 1

SESSION IS END. THANK YOU,

In []: